Standard Operating Procedure

Electric Strike & Electromagnetic Lock Installation

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Related Documents	SOP-LV-006, SOP-LV-014, SOP-LV-016
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Owner	Apex – Operations
Approved By	______________________________

NOTE ON THIS DOCUMENT:
Fail-safe/fail-secure and fire-door requirements below reflect general industry practice as documented by major hardware manufacturers (Camden, SDC, Allegion) and standard fire-door principles. Every fire-rated door assembly has its own listing; always confirm strike/lock selection against that door's specific fire label and the AHJ before installation. For life-safety egress interlock requirements (fire alarm release conditions), see SOP-LV-014 — that procedure governs regardless of which lock type is used here.

1. Purpose

This procedure establishes standard practices for selecting, installing, and testing electric strikes and electromagnetic locks (maglocks) on access-controlled doors.

2. Scope

Applies to all Apex technicians installing electric strikes or electromagnetic locks as part of an access control system. Life-safety fire alarm interlock requirements for these devices are covered separately in SOP-LV-014 and apply in addition to this procedure wherever relevant.

3. Fail-Safe vs. Fail-Secure — Selection

- Fail-secure (power-to-unlock): the door remains locked when power is removed. Voltage must be applied to unlock. This is the default choice for most electric strike applications and is required wherever the door must remain secure during a power outage.
- Fail-safe (power-to-lock): the door unlocks when power is removed. Used only where free egress/entry during a power failure is required by the application — this is a less common electric strike configuration and should not be assumed as default.
- Electromagnetic locks (maglocks) are fail-safe only — there is no fail-secure maglock. A maglock always releases when power is removed.
- An electric strike replaces the door's strike plate and only secures from the outside — the mechanical lockset (lever, panic bar) still allows manual egress from the inside regardless of the strike's powered state, unlike a maglock, which secures both sides of the door. This distinction is central to which device an application actually needs — do not select a maglock where an electric strike would meet the requirement, since a maglock always requires an additional REX device and full fire-alarm interlock per SOP-LV-014.

4. Fire-Rated Door Requirements

STOP — CONFIRM BEFORE PROCEEDING:
Electric strikes on fire-rated door assemblies must be fail-secure and UL (or WHI) listed for that specific fire rating. Fail-safe strikes are not permitted on fire-rated openings under a UL listing. Confirm the door's fire label and the strike's listing compatibility before ordering hardware, not at install time.

- The frame prep for an electric strike is larger than a standard mechanical strike prep — this cutout must be accounted for in the door assembly's fire listing. Do not field-modify a fire-rated frame beyond what the listed prep allows.

- A fire-rated door must positively latch in both the powered and unpowered state — verify this specifically as part of testing (Section 7), not just that the door "closes."
- Maglocks are not typically used on fire-rated doors in a way that maintains the door in an unlatched/held-open state, since fire doors must self-close and positively latch — confirm any maglock-on-fire-door application with the Site Supervisor before proceeding; this is not a default configuration.

5. Voltage & Duty Rating

- Common voltages are 12VDC and 24VDC — confirm which the specific strike/lock is rated for and which the access control panel's lock output supplies before wiring; do not assume.
- Electric strikes are rated for continuous duty (can remain energized indefinitely — typically fail-safe) or intermittent duty (energized only briefly to unlock — typically fail-secure). Confirm the duty rating matches how the door will actually be used; an intermittent-duty strike held energized continuously can overheat and fail prematurely.

6. Installation — Electric Strike

- Confirm the strike is compatible with the existing lockset (cylindrical, mortise, or exit device) — manufacturers publish compatible lockset lists; do not assume a strike will work with an arbitrary lockset.
- Mount the strike using the manufacturer's supplied fasteners into the prepped frame opening.
- Adjust the keeper for proper latch engagement — the deadlatch must be held in its intended depressed state by the correct faceplate, or the latch can be manipulated to open the door without triggering the strike. Use the specific faceplate the manufacturer specifies for the lockset in use.
- Wire MOV/surge suppression directly across the strike coil per the manufacturer's instructions, at the strike — not only at the control panel end — to protect the relay driving it from inductive kickback.

7. Installation — Electromagnetic Lock

- Mount the magnet body to the frame and the armature plate to the door per the manufacturer's alignment specifications — misalignment reduces holding force significantly even if the lock appears to engage.
- A REX device is required on every maglock installation — see SOP-LV-016. A maglock without a functioning REX path is a life-safety and code violation, not just a convenience gap.
- Confirm the fire alarm interlock requirements of SOP-LV-014 apply and are addressed before energizing any maglock on an egress door — do not commission a maglock ahead of its fire alarm interlock.
- Wire MOV/surge suppression directly across the magnet coil at the lock, per manufacturer instructions.

8. Testing

- Test the strike or lock in both locked and unlocked states — confirm the latch/armature engages fully when locked and releases cleanly when commanded to unlock.
- Test the fail-safe or fail-secure behavior specifically: disconnect power and verify the door behaves as designed (locks or unlocks, matching the intended configuration) — do not assume the factory default matches the application without testing it.
- For fire-rated assemblies: with power removed, confirm the door still self-closes and positively latches.
- For maglocks: confirm REX activation releases the lock, confirm the fire alarm interlock releases the lock (per SOP-LV-014), and confirm manual free egress via any required push-button/crash bar.
- Record the fail-safe/fail-secure configuration, voltage, and duty rating on the site door schedule/configuration sheet.

9. References

- Manufacturer installation instructions for the specific strike/lock model in use (govern where more restrictive than this SOP)
- UL 294 — Access Control System Units
- NFPA 80 — Standard for Fire Doors and Other Opening Protectives
- SOP-LV-006 — Utility Strike Prevention
- SOP-LV-014 — Fire Alarm Interface & Life-Safety Interlock
- SOP-LV-016 — REX & Door Position Switch Installation